

netweevil QGIS Plugin

This plugin talks directly to the running **netweevil** API.

It does not shell out to the CLI. QGIS sends JSON requests to the API, receives JSON or GeoJSON responses, and loads the returned geometries into the map immediately.

What It Does

- connects to a running **netweevil api serve** instance
- reads the loaded dataset and available profiles from **/v1/service**
- lets you pick route start and end points directly from the QGIS map canvas
- can pull route points from a selected point feature in the active layer
- submits route, OD, matrix, and service-area requests to the API
- loads route, OD, matrix, and service-area geometries into QGIS from API responses
- can build matrix origins and destinations from loaded QGIS point layers
- can build service-area origins from map picks, selected point features, or the current route endpoints
- exposes disconnected-network policies and opt-in degraded-routing controls for route and batch requests
- supports both normal JSON responses and direct GeoJSON responses
- remembers connection and analysis settings between QGIS sessions

Install In QGIS

1. Start the API:

```
cargo run -p netweevil-cli -- api serve \  
--dataset ile_de_france_2026_03 \  
--default-profile examples/profiles/car_research_v1.yml \  
--profile examples/profiles/pedestrian_research_v1.yml \  
--bind 127.0.0.1:8080
```

2. Copy or symlink **netweevil_qgis** into your QGIS profile plugin directory.

Linux example:

```
mkdir -p ~/.local/share/QGIS/QGIS3/profiles/default/python/plugins  
ln -s /home/elmeriniemi/stuff/netweevil/qgis_plugin/netweevil_qgis ~/.local/share/QGIS/QGIS3/profiles/default/python/plugins
```

Windows QGIS 4 example:

```
New-Item -ItemType Directory -Force "$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins"  
cmd /c mklink /D "$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins\netweevil_qgis" "%cd%\netweevil_qgis"
```

3. In QGIS, enable the plugin from **Plugins -> Manage and Install Plugins**.

First Run

Set these fields in the dock:

- **Workspace root:** local root used only for browsing and saving request/response files
- **API base URL:** for example `http://127.0.0.1:8080`
- **Timeout seconds**
- **Response format:** JSON or GeoJSON
- **Profile:** either the service default or any profile preloaded by the API

Then press **Refresh Service**.

The plugin will show:

- the dataset currently loaded by the API
- the API default profile
- the available profile list
- the loaded dataset bounds when available

Windows QGIS + WSL

This is simpler now than the old CLI integration. Run the API wherever **netweevil** lives, then point QGIS at that HTTP endpoint.

Examples:

- API inside WSL, QGIS on Windows:
 - **API base URL:** `http://127.0.0.1:8080` if you expose it locally from WSL
 - **Workspace root:** any Windows-accessible folder you want to use for request and response files
- Native Linux QGIS:
 - **API base URL:** `http://127.0.0.1:8080`
 - **Workspace root:** `/home/elmeriniemi/stuff/netweevil`

Example Flows

Route

1. Open the plugin dock.
2. Set **Workspace root** and **API base URL**.
3. Press **Refresh Service**.
4. Choose either the service default profile or a specific loaded profile.
5. In the **Route** tab, keep the generated route id or set your own.
6. Click **Pick Start**, then click on the map canvas.
7. Click **Pick End**, then click on the map canvas.
8. Optionally use **From Selected Feature** if you already have a point selected in the active point layer.

9. Choose the returned route detail you want, including segment rows, road-type totals, and surface totals.
10. Optionally press **Save Request** if you want the route JSON on disk.
11. Press **Run Route**.

The plugin loads the route into a grouped set of QGIS layers and tables automatically. When the API returns them, the group includes the main route, segment rows, hop segments, violations, road-type breakdowns, and surface breakdowns. After a successful run, the default route id advances so the next query does not overwrite the previous route name.

OD

Use:

- pairs file: `examples/requests/od_pairs.csv`
- response path: `.netweevil/runs/qgis-od.geojson`

Then press **Run OD**.

Matrix

Use either loaded point layers or files:

- set **Origins and Destinations** source to **Loaded point layer** to build the matrix from QGIS layers already in the project
- optionally choose an ID field and **Use selected features only**
- or switch either side to **CSV or JSON file**
- set response path: `.netweevil/runs/qgis-matrix.geojson`

Then press **Run Matrix**.

Service Area

Use the dedicated **Service Area** tab:

1. Set an **Analysis id**.
2. Enter one threshold list such as 300, 600.
3. Pick either **Distance (m)** or **Travel time (s)** for the whole threshold list.
4. Append origins with **Pick On Map**, **Add Selected Features**, **Use Route Start**, or **Use Route Start + End**.
5. Choose **Output mode**, **Band mode**, **Boundary mode**, and **Multi-origin mode**.
6. Optionally expand **Connectivity Controls** to set disconnected-network behavior.
7. Press **Run Service Area**.

The plugin saves the raw API response and loads grouped sublayers back into QGIS by threshold and geometry type.

Advanced Route And Batch Controls

- **Route** exposes a dedicated **Returned route detail** section for geometry mode, segment rows, and route breakdown request flags.
- **Route**, **Batch**, and **Service Area** keep disconnected-network policies under collapsible connectivity controls.
- Unsafe degraded-routing options remain under a separate collapsed **Unsafe Failure Modes** section and are off by default.
- The plugin shows a preflight confirmation before sending a request with unsafe failure modes enabled.
- Diagnostics now include suggested next actions in the dock log when the API returns them.

Notes

- The plugin does not embed the Rust engine; it calls the running HTTP API.
- Relative request/input/output paths are resolved against the configured workspace root.
- OD and matrix CSV inputs are parsed locally and sent to the API as JSON.
- Route points are transformed from the current map or layer CRS into WGS84 before being sent to the API.
- YAML request files are not supported by the plugin in API mode.
- Route detail layers use JSON responses internally when needed, even if the dock is set to **GeoJSON**, so the plugin can still load segment rows and breakdown tables.
- When **Response format** is **JSON**, the plugin still builds temporary GeoJSON layers locally when geometry is present in the API response.
- Service-area outputs are loaded into grouped sublayers so thresholds and geometry modes stay inspectable in QGIS.
- In **WSL** mode, output layers are loaded back into QGIS through `\\wsl$\\<distro>\\. . . .`